

Project 3: Non-linear light response curves

Overview: Here, you will implement a Bayesian non-linear regression model to evaluate how plant photosynthesis relates to light level. Leaf-level photosynthesis rates are tightly coupled to the amount of light reaching the leaf surface, and the increase in net photosynthesis rate (A_n) as a function of light is described by a “light response curve.” Different species and plants growing under different conditions (e.g., high light versus shade) often have different light response curves.

Important features of a light-response curve are:

- (1) the initial slope of the curve at $PAR = 0$ represents the light-use efficiency (LUE);
- (2) as A_n increases with PAR , it reaches a maximum value that approaches $A_n = A_{\max} - R_d$, thus $A_{\max}$ is the maximum gross photosynthesis rate;
- (3) the y-intercept, which represents the dark respiration rate, R_d .

The data: Light response curves were measured on 21 seedlings from two different species (sugar maple vs. black oak) and under two different canopy conditions (gap vs. shade). The “gap” conditions are characterized by exposure to high light; the “shade” conditions generally reduce the amount of light reaching the seedlings due to overhanging canopies of nearby trees. Photosynthesis measurements were made under different light levels, ranging from 2000 to 0 $\mu\text{mol m}^{-2} \text{s}^{-1}$ in 9 steps, resulting in 10 observations per seedling or “curve” (for $N = 210$ observations).

The model: Assume the A_n data are normally distributed around a mean, μ , and model the mean as a non-linear function of PAR via a rectangular hyperbola equation:

$$A_{n_i} \sim \text{Normal}(\mu_i, \tau)$$

$$\mu_i = \frac{\theta_{1,c(i)} \cdot PAR_i \cdot \theta_{2,c(i)}}{\theta_{1,c(i)} \cdot PAR_i + \theta_{2,c(i)}} - \theta_{3,c(i)}$$

The parameters (θ 's) are indexed by $c(i)$, which denotes curve c ($c = 1, 2, \dots, 21$) associated with observation i . Assign hierarchical priors to the curve-level θ 's, treating curves as nested in species (2 levels; sugar maple [1] vs. black oak [2]) and canopy type (2 levels; gap [1] vs shade [2]). Thus, curves (or seedling) are treated as a random effect and species and canopy type as fixed effects.

Assign relatively non-informative priors to all root nodes, including the species- and canopy-type-level parameters, and all precision terms.

Specifically, assign hierarchical priors to the curve-level parameters (θ) that directly obey their domains, whereby $A_{\max} > 0$, $LUE > 0$, and $R_d > 0$. Thus, assign hierarchical normal priors to the log-scale parameters, $\log(A_{\max})$, $\log(R_d)$, $\log(LUE)$, treating curves as nested in species and canopy type. That is, define $\phi = \log(\theta)$ and assign:

$$\phi_{k,c} \sim \text{Normal}(\mu_{\phi_{k,s(c),t(c)}}, \tau_{\phi_k}) \quad \text{for } k = 1, 2, 3$$

$$\theta_{k,c} = \exp(\phi_{k,c})$$

Where $s(c)$ and $t(c)$ denote species s ($s = 1, 2$) and canopy type t ($t = 1, 2$) associated with curve c . Finally, assign conjugate, relatively non-informative priors to all root nodes, including each μ_ϕ term and all precisions (τ_ϕ terms and τ). Note that the population-level means (μ_ϕ terms) vary by species and canopy type, but the population-level precision terms (τ_ϕ) are shared across species and canopy type and only vary by parameter type k .

Specify ballpark initials for the root node log-scale photosynthesis parameters (i.e., μ_ϕ terms) and all precision terms. Run the model for 3 chains, for 20000 iterations per chain. Evaluate mixing and convergence, and compute posterior statistics. Is there evidence that the light-response characteristics (parameters) differ among species and/or canopy types?